

REAL-TIME OBJECT DETECTION USING HARDWARE-ACCELERATED CNN ON XILINX ZYNQ FPGA WITH ARM PROCESSOR

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ABSTRACT

- This project aimed to design and implement a hardware-accelerated Convolutional Neural Network (CNN) inference system on a Xilinx Zynq System-on-Chip (SoC). The objective was to offload computationally intensive CNN operations to the FPGA fabric (Programmable Logic - PL) while managing preprocessing and control through the Processing System (PS).
- We design leveraged AXI interfaces and AXI DMA for high-speed data transfer between PS and PL. The CNN accelerator was developed using Vitis HLS and integrated into Vivado Block Design. Although full hardware validation could not be completed due to AXI-DMA and synchronization challenges, substantial progress was made in hardware design, integration, simulation, and debugging.

INTRODUCTION

- Deep learning models, particularly CNNs, demand high computational throughput. Running CNN inference purely on a processor results in high latency and power consumption.

The Xilinx Zynq SoC provides:

- ARM Cortex-A9 Processing System (PS)
- FPGA Programmable Logic (PL) – we used PYNQ Z2
- High-performance AXI interconnects

OBJECTIVE

1. Design CNN inference module using Vitis HLS
2. Implement AXI4-Stream based hardware accelerator
3. Integrate accelerator with AXI DMA in Vivado
4. Transfer data between PS and PL using DMA
5. Evaluate hardware utilization (LUTs, BRAM, DSP)
6. Optimize latency and throughput
7. Compare PS-only vs PL-accelerated execution

SYSTEM ARCHITECTURE FOR PYNQ Z2

The system consists of:

- Processing System (ARM Cortex-A9)
- AXI DMA Engine
- CNN Hardware Accelerator (HLS IP)
- AXI Interconnect
- DDR Memory

Data Flow:

1. Input image stored in DDR (PS side)
2. DMA MM2S transfers input to CNN IP
3. CNN accelerator processes data
4. Output streamed back via S2MM
5. Result stored in DDR

METHODOLOGY

- **IP Core Generation**–Synthesized the design and exported it as a custom hardware IP for integration in Vivado.
- **Vivado Block Design Integration**–Integrated the CNN IP with Zynq Processing System, AXI DMA, AXI Interconnect, clock, and reset modules.
- **AXI DMA Configuration**–Configured MM2S and S2MM channels for high-speed data transfer between PS (DDR memory) and PL accelerator.
- **Bitstream Generation and Deployment**–Generated the FPGA bitstream and deployed it on the target board using PYNQ environment.
- **Hardware Validation and Debugging**–Attempted end-to-end validation of PS–PL data transfer; encountered AXI-DMA synchronization issues and performed systematic debugging of clock, reset, and stream configurations.

METHODOLOGY

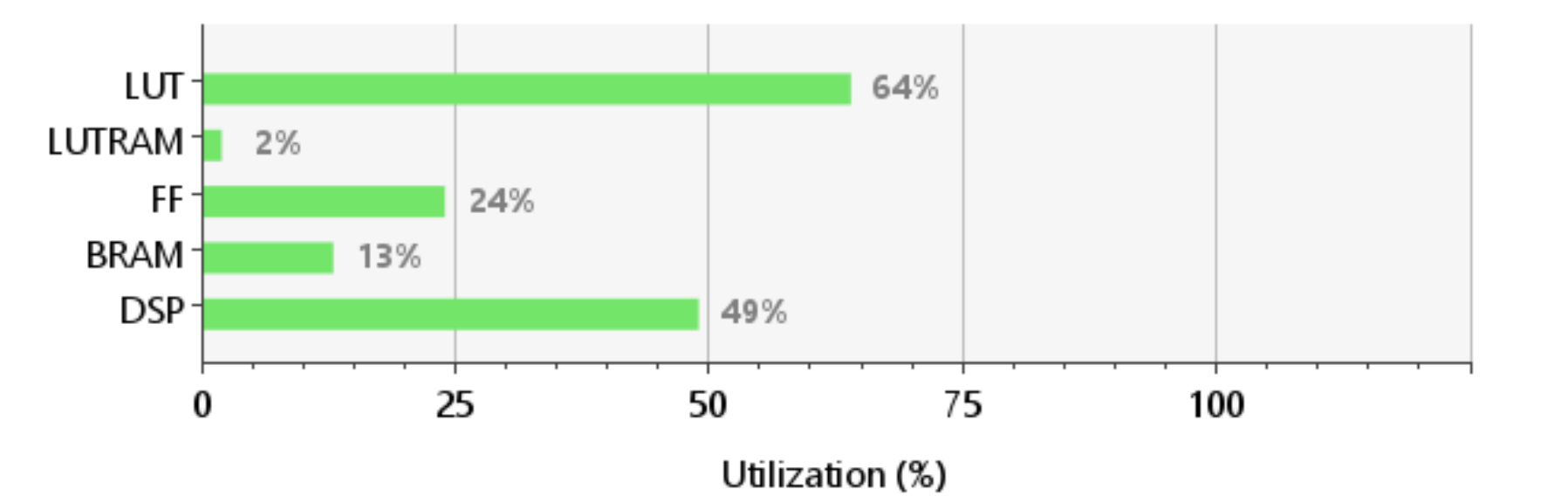
- **Problem Definition and System Planning**–Defined the objective to implement a CNN inference accelerator on Xilinx Zynq SoC using hardware–software co-design, offloading computation to Programmable Logic (PL).
- **CNN Algorithm Development**–Implemented convolution, activation (ReLU), and fully connected layers in C/C++ for hardware synthesis.
- **High-Level Synthesis (HLS) Design**–Converted the CNN algorithm into hardware using Vitis HLS and defined AXI4-Stream and AXI-Lite interfaces.
- **Hardware Optimization Techniques**–Applied loop pipelining and loop unrolling to improve throughput and reduce latency while analyzing resource trade-offs.
- **Functional Simulation (C-Simulation)**–Verified correctness of CNN inference at the behavioral level before hardware synthesis.

Design Validation & Resources

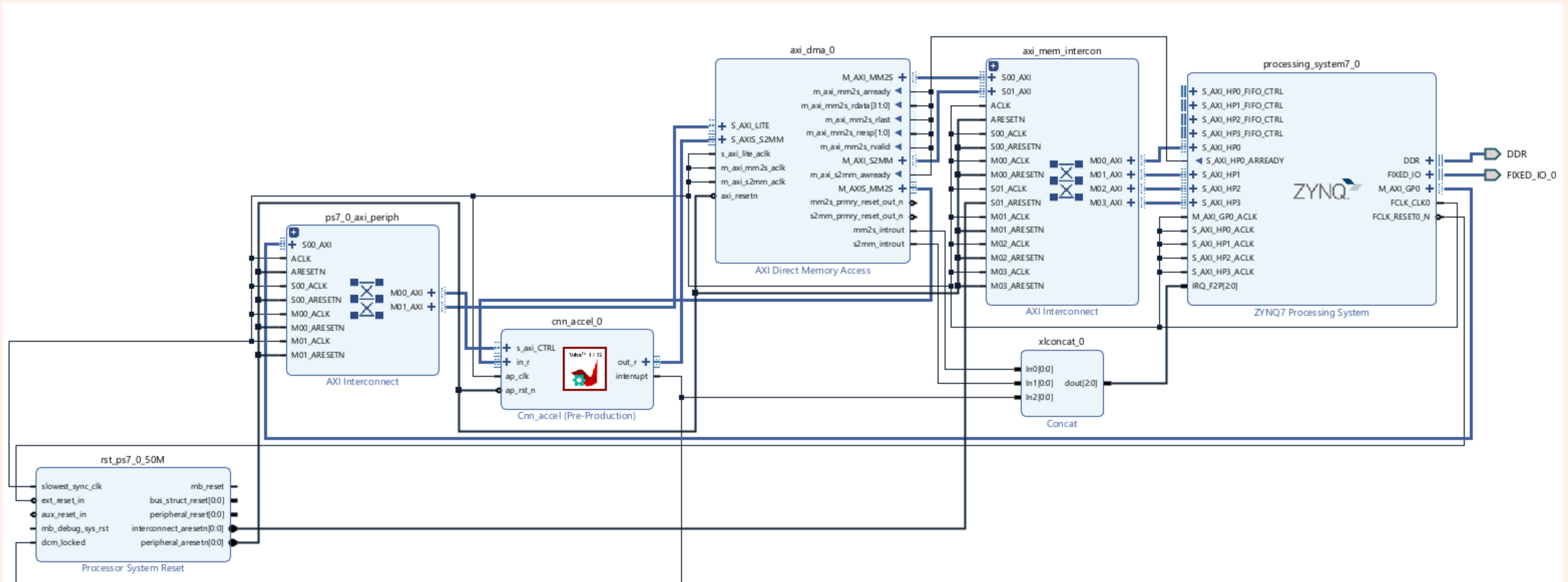
Utilization of PYNQ Z2

Name ¹	Slice LUTs (53200)	Slice Registers (106400)	F7 Muxes (26600)	F8 Muxes (13300)	Slice (13300)	LUT as Logic (53200)	LUT as Memory (17400)	Block RAM Tile (140)	DSPs (220)	Bonded IOPADs (130)	BUFGCTRL (32)
▼ N design_1_wrapper	27482	21689	262	85	8934	27236	246	18.5	108	130	1
> I design_1_i (design_1)	27482	21689	262	85	8934	27236	246	18.5	108	0	1

Resource	Utilization	Available	Utilization %
LUT	34101	53200	64.10
LUTRAM	327	17400	1.88
FF	25966	106400	24.40
BRAM	18.50	140	13.21
DSP	108	220	49.09



Hardware Design Overview



RESULT

- Worst Negative Slack (WNS): 1.939 ns. This is a positive value, indicating that the design comfortably meets setup requirements.
- Total Negative Slack (TNS): 0.000 ns. This confirms there are no paths failing setup constraints.
- Worst Hold Slack (WHS): 0.045 ns. While this is a thin margin, it is positive, meaning the design passes hold timing.
- Total Hold Slack (THS): 0.000 ns. No paths are violating hold requirements.
- Worst Pulse Width Slack (WPWS): 3.750 ns. The clock signal maintains its shape well within the required limits.
- Total Pulse Width Negative Slack (TPWS): 0.000 ns.

CONCLUSION

- This project aimed to design and integrate a CNN hardware accelerator on the Xilinx Zynq SoC platform using Vitis HLS and Vivado. The CNN accelerator was successfully developed, synthesized, and integrated with AXI DMA and AXI interconnect infrastructure. Bitstream generation and PS-PL communication setup were completed, and functional verification at the simulation level was achieved.
- During full system validation, AXI-DMA synchronization issues prevented stable end-to-end inference execution within the project timeline. Despite this, the project successfully demonstrated hardware accelerator design methodology, AXI-based system integration, and hardware/software co-design principles.
- The work provided strong practical exposure to FPGA-based AI acceleration and complex SoC debugging. With further refinement of DMA configuration and stream synchronization, the system can be fully operational and optimized in future development.

REFERENCE

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THANK YOU

For your attention

